

## Lecture 8 Tutorial — Answers

### Q1. Non-parametric Modeling

1. underlying **densities**
2. **multimodal**

### Q2. Pros and Cons

1. **T**
2. **T**
3. **F**
4. **T**

### Q3. kNN Classification Rule

C. majority vote among the nearest neighbours' labels

### Q4. kNN Window / Neighbourhood Interpretation

The lecture describes a cell centred at  $x$  that grows until it captures  $k_n$  samples.

1. They are called the  $k_n$  **nearest neighbours**.
2. If the window is too small: **most of the volume may be empty / over-sensitive to noise**.
3. If the window is too large: **important variations may be lost**.

### Q5. kNN Computation

Use Euclidean distance on (Temperature, Humidity) only for the query (74, 82).

$$d = \sqrt{(T - 74)^2 + (H - 82)^2}$$

| Row | $(T, H)$ | Distance to $(74, 82)$      | Play  |
|-----|----------|-----------------------------|-------|
| 1   | (85,85)  | $\sqrt{130} \approx 11.402$ | maybe |
| 2   | (80,90)  | $\sqrt{100} = 10.000$       | no    |
| 3   | (83,86)  | $\sqrt{97} \approx 9.849$   | yes   |
| 4   | (70,96)  | $\sqrt{212} \approx 14.560$ | maybe |
| 5   | (68,80)  | $\sqrt{40} \approx 6.325$   | yes   |
| 6   | (65,70)  | $\sqrt{225} = 15.000$       | no    |
| 7   | (64,65)  | $\sqrt{389} \approx 19.723$ | yes   |
| 8   | (72,95)  | $\sqrt{173} \approx 13.153$ | no    |
| 9   | (69,70)  | $\sqrt{169} = 13.000$       | yes   |
| 10  | (75,80)  | $\sqrt{5} \approx 2.236$    | maybe |
| 11  | (75,70)  | $\sqrt{145} \approx 12.042$ | yes   |
| 12  | (72,90)  | $\sqrt{68} \approx 8.246$   | maybe |
| 13  | (81,75)  | $\sqrt{98} \approx 9.899$   | yes   |
| 14  | (71,91)  | $\sqrt{90} \approx 9.487$   | no    |

1. Distances are listed in the table above.
2. The 3 nearest neighbours are rows **10, 5, 12**.
3. Their labels are maybe, yes, maybe, so the majority vote gives **maybe**.

## Q6. Sensitivity to $k$

Using the neighbour ranking:

10, 5, 12, 14, 3, 13, 2, 1, 11, 9, 8, 4, 6, 7

1. Predicted label for  $k = 3$ : **maybe**
2. Predicted label for  $k = 5$ : **maybe**
3. Predicted label for  $k = 7$ : **yes**

## Q7. Decision Tree Stopping Conditions

B. majority vote (mode) of the labels in the node

## Q8. Entropy

1.  $p_k$  is the proportion of class  $k$  in  $D$ .
2.  $D_v$  is the subset of  $D$  with attribute value  $v$  for **attribute**  $x_j$ .

## Q9. Entropy Calculation (3-class Node)

Given

$$\text{Yes} = 6, \quad \text{No} = 5, \quad \text{Maybe} = 3, \quad |D| = 14$$

So

$$p_{\text{Yes}} = \frac{6}{14}, \quad p_{\text{No}} = \frac{5}{14}, \quad p_{\text{Maybe}} = \frac{3}{14}$$

1.

$$\text{entropy}(D) = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{5}{14} \log_2 \frac{5}{14} - \frac{3}{14} \log_2 \frac{3}{14} \approx 1.531$$

2. The mode is **Yes**.

## Q10. Information Gain for a Split

Parent entropy:

$$\text{entropy}(D) \approx 1.531$$

For

$$D_1 : (2, 3, 1), \quad |D_1| = 6$$

$$D_2 : (3, 0, 1), \quad |D_2| = 4$$

$$D_3 : (1, 2, 1), \quad |D_3| = 4$$

1.

$$\text{entropy}(D_1) = -\frac{2}{6} \log_2 \frac{2}{6} - \frac{3}{6} \log_2 \frac{3}{6} - \frac{1}{6} \log_2 \frac{1}{6} \approx 1.459$$

$$\text{entropy}(D_2) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \approx 0.811$$

$$\text{entropy}(D_3) = -\frac{1}{4} \log_2 \frac{1}{4} - \frac{2}{4} \log_2 \frac{2}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 1.500$$

2. Weighted entropy:

$$\frac{6}{14}(1.459) + \frac{4}{14}(0.811) + \frac{4}{14}(1.500) \approx 1.286$$

3.

$$\text{Gain}(D, x_j) = 1.531 - 1.286 = 0.245$$

## Q11. Gain Ratio

Using split sizes 6, 4, 4 out of 14:

1.

$$\text{IV} = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{4}{14} \log_2 \frac{4}{14} - \frac{4}{14} \log_2 \frac{4}{14} \approx 1.557$$

2.

$$\text{GainRatio}(D, x_j) = \frac{0.250}{1.557} \approx 0.161$$

## Q12. Clustering

### Part A — Agglomerative (Hierarchical) Clustering

Using single linkage and the city-block distances:

1. The first merge is  $C$  **and**  $E$ , since

$$d(C, E) = 2$$

is the smallest distance.

2. The second merge is  $A$  **with**  $(C, E)$ , because

$$d(A, C) = 3, \quad d(A, E) = 13,$$

so by single linkage

$$d(A, \{C, E\}) = \min(3, 13) = 3.$$

3. The clusters after the second merge are:

$$\{A, C, E\}, \quad \{B\}, \quad \{D\}.$$

## Part B — One Iteration of k-means

Points:

(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)

Initial centroids:

$$c_1^{(0)} = (1.0, 1.0), \quad c_2^{(0)} = (5.0, 7.0)$$

1. Cluster assignment after one iteration:

$$(1.0, 1.0) \rightarrow C_1$$

$$(1.5, 2.0) \rightarrow C_1$$

$$(3.0, 4.0) \rightarrow C_1$$

$$(5.0, 7.0) \rightarrow C_2$$

$$(3.5, 5.0) \rightarrow C_2$$

$$(4.5, 5.0) \rightarrow C_2$$

$$(3.5, 4.5) \rightarrow C_2$$

So,

$$C_1 = \{(1.0, 1.0), (1.5, 2.0), (3.0, 4.0)\}$$

$$C_2 = \{(5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)\}$$

2. Updated centroids:

$$c_1^{(1)} = \left( \frac{1.0 + 1.5 + 3.0}{3}, \frac{1.0 + 2.0 + 4.0}{3} \right) = \left( \frac{5.5}{3}, \frac{7}{3} \right) \approx (1.833, 2.333)$$

$$c_2^{(1)} = \left( \frac{5.0 + 3.5 + 4.5 + 3.5}{4}, \frac{7.0 + 5.0 + 5.0 + 4.5}{4} \right) = \left( \frac{16.5}{4}, \frac{21.5}{4} \right) = (4.125, 5.375)$$